

空间飞行物拟合曲线

$$x = -2.29 \times 10^{-5} t^4 - 0.007051 t^3 - 3.222 t^2 - 1888 t - 2.147 \times 10^5$$

$$y = 8.367 \times 10^{-6} t^4 - 1.993 \times 10^{-5} t^3 + 0.9286 t^2 + 165.2 t + 6.729 \times 10^6$$

$$z = 6.477 \times 10^{-5} t^4 - 0.003299 t^3 + 6.887 t^2 + 643.2 t + 2.103 \times 10^6$$

$$\dot{x} = -9.16 \times 10^{-5} t^3 - 0.021153 t^2 - 6.444 t - 1888$$

$$\dot{y} = 3.3468 \times 10^{-5} t^3 - 5.979 \times 10^{-5} t^2 + 1.8572 t + 165.2$$

$$\dot{z} = 2.5908 \times 10^{-4} t^3 - 0.009897 t^2 + 13.774 t + 643.2$$

$$\ddot{x} = -2.748 \times 10^{-4} t^2 - 0.042306 t - 6.444$$

$$\ddot{y} = 1.00404 \times 10^{-4} t^2 - 1.1958 \times 10^{-4} t + 1.8572$$

$$\ddot{z} = 7.7724 \times 10^{-4} t^2 + 0.019794 t + 13.774$$